Worksheet 3

Dispersion and concentration – DETAILED SOLUTIONS

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Detailed solution – Exercise 1

$N = 20$, $\sum n_i x_i = 165$, so $\bar{x} = 8.25$. The detailed table gives $\sum n_i |x_i - \bar{x}| = 79.5$ and $\sum n_i (x_i - \bar{x})^2 = 389.75$.

$$e = 79.5/20 = \boxed{3.975}, \quad V = 389.75/20 = \boxed{19.4875},$$

$$\sigma = \boxed{4.414}, \quad \text{range} = 15 - 2 = 13,$$

$$CV = 4.414/8.25 = \boxed{53.5\%}.$$

Detailed solution – Exercise 2

Using class midpoints, $\sum n_i c_i = 6665$ and $\sum n_i c_i^2 = 227036.5$.

$$\bar{x} = 31.7381,$$

$$V = 227036.5/210 - (31.7381)^2 = \boxed{73.8195},$$

$$\sigma = \boxed{8.592 \text{ euros}}, \quad CV = \boxed{27.07\%}.$$

With $Q_1 = 26.56$ and $Q_3 = 37.50$,

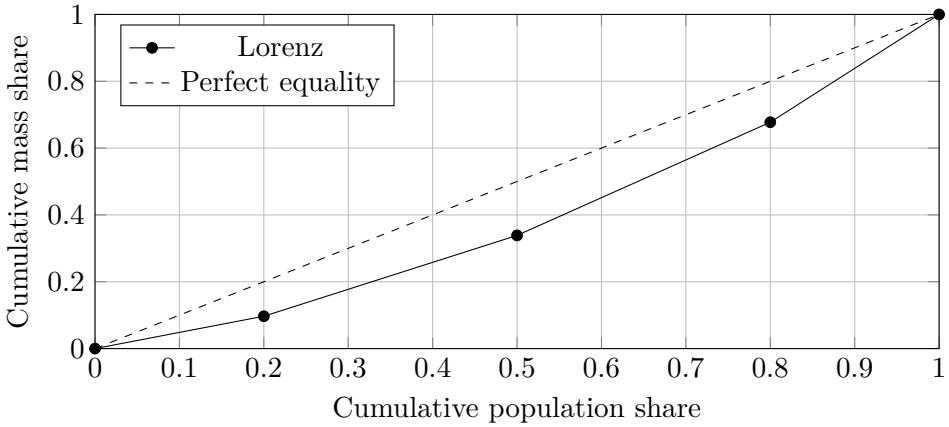
$$IQR = \boxed{10.94 \text{ euros}}.$$

Detailed solution – Exercise 3

Midpoints are $15, 25, 35, 50$. Masses are $300, 750, 1050, 1000$, so total mass $S = 3100$. Cumulative population shares: $20\%, 50\%, 80\%, 100\%$; cumulative mass shares: $9.68\%, 33.87\%, 67.74\%, 100\%$. Half the total mass is 1550. It lies in $[30, 40[$:

$$MI = 30 + 10\frac{1550 - 1050}{1050} = \boxed{34.76}.$$

Lorenz points are $(0, 0), (0.2, 0.0968), (0.5, 0.3387), (0.8, 0.6774), (1, 1)$.



Trapezoid area $A = 0.39516$, so

$$G = 1 - 2A = \boxed{0.20968 = 20.97\%}.$$